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In [1]: import numpy as np, cv2, os, matplotlib.pyplot as plt
from sklearn.datasets import fetch_olivetti_faces
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
path = r"C:\Users\mech\Desktop\PHY\olivetti_faces.npy"
imgs, labels = (np.load(path), np.array([i//10 for i in range(len(np.load(path))
    if os.path.exists(path) else (lambda d:(d.images,d.target))(fetch_olivetti_face
print("Dataset:", imgs.shape, "| Classes:", len(np.unique(labels)))
imgs = np.array([cv2.cvtColor((i*255).astype("uint8"), cv2.COLOR_GRAY2BGR) for i
features = np.array([np.hstack([
    cv2.normalize(cv2.calcHist([im],[c],None,[256],[0,256]),None).flatten()
    for c in range(3)]) for im in imgs])
print("Feature Shape:", features.shape)
Xtr,Xte,Ytr,Yte,Imtr,Imte = train_test_split(
    features,labels,imgs,test_size=0.2,random_state=42,stratify=labels)
model = RandomForestClassifier(n_estimators=200,random_state=42).fit(Xtr,Ytr)
pred = model.predict(Xte)
print("Accuracy:", round(accuracy_score(Yte,pred)*100,2), "%")
plt.figure(figsize=(10,8))
for i in range(6):
    plt.subplot(2,3,i+1)
    plt.imshow(cv2.cvtColor(Imte[i],cv2.COLOR_BGR2RGB))
    plt.title(f"A:{Yte[i]} | P:{pred[i]}")
    plt.axis("off")
plt.suptitle("Color Face Recognition Output")
plt.tight_layout();plt.show()
```

downloading Olivetti faces from <https://ndownloader.figshare.com/files/5976027> to
C:\Users\CAD LAB 03\scikit_learn_data
Dataset: (400, 64, 64) | Classes: 40
Feature Shape: (400, 768)
Accuracy: 77.5 %

Color Face Recognition Output

A:1 | P:1



Color Face Recognition Output

A:35 | P:26



Color Face Recognition Output

A:18 | P:18



Color Face Recognition Output

A:16 | P:16



Color Face Recognition Output

A:39 | P:37



Color Face Recognition Output

A:23 | P:23



In []: SE-A S3
Roll no -338